



Citi Bike × Insurance: Safety + Support + Optional Protection

Comprehensive Analysis of Citi bike usage pattern across NYC and Jersey city combining trip data and risk proximity metrics.



Problems

1. What problem are we solving for each side ?

- Citi Bike riders have no built-in coverage; incidents create support burden + reputational risk [citybikenyc](#), [DavidoffLow](#)
- Operators (lyft) need safety prioritization and a shock playbook (COVID-like disruptions) [Plos](#), [PubMedCent](#), [DavidoffLow](#)
- Citi Bike and lyft are [facing with rebalancing failure](#), [Ghost Dock](#) and No Public Subsidy. [ibo](#)
- Insurers need controlled distribution and privacy-safe verification (no heavy data sharing) .

2. What are the 2–3 concrete products we can ship in a pilot?

- what appears in the app and what happens after an incident

3. How do we de-risk operationally, legally, and reputationally?

- proof-of-ride token,
- no dynamic pricing,
- safety framing

4. How do we use data responsibly (minimal data, maximal learning)?

5. How do we measure success fast (and kill it if it harms the core business)?

- We need KPIs + thresholds (“kill switches”)

Business Value

Citi Bike / Operator

-  safety outcomes
-  reputation
-  downtime
-  support burden
-  city relationship
-  revenue with low UX risk
-  play book disruptions

Insurer (e.g., AXA)

-  scalable distribution
-  controable loss ratio
-  fraud controls
-  compliant claims ops
-  embedded insurance¹

City / Public

-  safety investment focus
-  resilience plan
-  Plan

¹ Expected to grow by 2032 and over 30% of all insurance transactions likely occurring through embedded channels by 2028 [BCG](#), [Zurich Insurance](#)

Ideas could lead us to get there?

4-Zons

1. High Risk/ High Exposure
2. High Risk/ Low Exposure
3. Low Risk/ High Exposure
4. Low Risk/ Low Exposure

Time Pattern

- 1.Year
- 2.Month
- 3.Week
- 4.Day
- 5.Hour

Any Disruption Mode



Partnership Ideas

1. **4-Zone Safety + Support + Optional Protection** (flagship “balanced” plan)
2. **Zone 1 First + Safety Fund Loop** (serious safety + city politics plan)
3. **Concierge-First Wedge, then Insurance** (internal champion plan)
4. **Zone 3 Growth Engine + Zone 1 Safety Guardrail** (revenue-first but safe)
5. **Corporate / Campus Bundle + 4-Zone Safety Reporting** (stable underwriting plan)

Partnership Plan 1 – 4-Zone Safety + Support + Optional Protection

What it is

-  Safety Nudges (In-App)
-  Incident Concierge (Help & Claim Link)
-  Optional Micro-Protection (Add-on at Checkout)

Zone Playbook

High/High = Risk/Exposure

 Zone 1 High/High Strongest Safety + + Concierge	⇒ Zone 2 High/Low Safety Nudges Only
 Zone 3 Low/High Offer Shown Most + Light Nudges	 Control Gow/Low + Experiments

Time-Pattern Operating Rules

- Peak Months: Increase Prompts + Staffing
- Wededay vs Weekend: Commuter vs Is Leisure Copy
- Hour-of-Day: Activate Peak / Risk Hours

Disruption Mode (COVID)

- Concierge Up, Sales Down, Keep Pricing Stable

Zone 1 First + Safety Fund Loop

— serious safety + city politics plan

What it is



A plan designed to win buy-in from city leadership and stakeholders who are allergic to "monetizing risk."

What ships (pilot)

- Zone 1 "Hotspot Pack": stronger safety nudges + micro-education cues
- Concierge 
- Optional add-on positioned as funding safety & rider recovery 

The Signature Mechanism: Safety Fund



A fixed % of premium goes into a visible Safety Fund.

Zone Playbook

- **Zone 1:** primary investment + reporting
- **Zone 2:** monitor + low-cost nudges
- **Zone 3:** limited sales prompts; keep optics clean
- **Zone 4:** ignore unless needed as control

Disruption Mode

- In disasters: Safety Fund focuses on support & safe riding guidance; sales prompts minimized.

Partnership Plan 3 — Concierge-First Wedge, then Insurance (internal champion plan)

What it is

You start with the feature that makes ops/support love you, then attach insurance once trust exists.



Phase 1 (2–4 weeks)

- ◆ Incident concierge across the system, most prominent in Zone 1 peak hours
- ◆ Proof-of-ride token logic behind the scenes (even if no insurance yet)



Phase 2 (weeks 5–12)

Add opt-in protection:

- ◆ shown mostly in Zone 3 (clean conversion)
- ◆ shown in Zone 1 with safety framing (not fear)

Zone Playbook

High/High = Risk/Exposure

Zone 1 High/High Concierge + safety cues first; insurance later and sotty	Zone 2 High/Low → Minimal
Zone 3 Low/High 💰 Insurance conversion engine	Zone 4 Low/Low 🔧 Minimal

Time patterns

Used to:

- ◆ staff concierge during peaks
- ◆ time the insurance launch (e.g., at start of high season)

Disruption Mode

- ◆ During shocks: concierge becomes the core product; insurance prompts reduced.

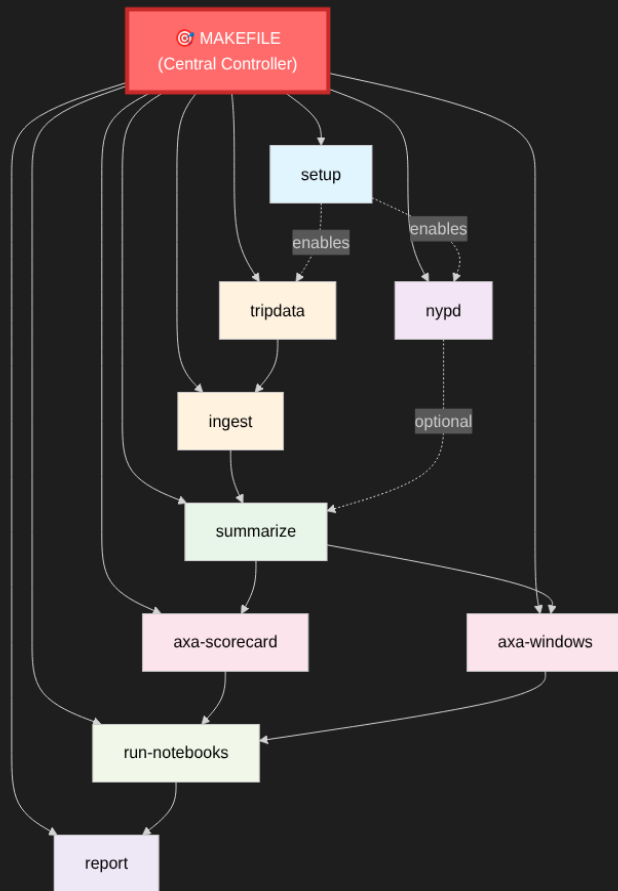


Technical Part

Data & Methodology

1. **Data sources:** Citi Bike trips; NYDP dataset; (JC: exposure only)
2. **Unit of analysis:** station-month (It can be changed based on Stakeholders expectations)
3. **Metrics:**
 - Exposure = touchpoints (trips)
 - Risk proxy = crashes near station / trips "EB-smoothed"
4. **Controls:**
 - min_trip threshold
 - Poisson CI
 - EB smoothing
 - Credibility
5. **Time slicing**
 - Year
 - Month
 - DOW
 - hour

How
This
Pipeline
is
working



~/Desktop/citibike-insurance-challenge



citibike-insurance-challenge

AXA • CitiBike • Empirical Bayes Risk Scoring Pipeline • NYC • Jersey City

```
citibike-insurance-challenge/
├── data/
│   ├── raw/
│   │   ├── citibike/ Downloaded ZIP files (by run)
│   │   └── nypd/ NYPD crash CSV (you provide)
│   └── processed/
│       ├── citibike_parquet/ Optimized trip data
│       └── nypd_crashes_*.csv Filtered crashes
├── summaries/ Analysis outputs
│   ├── yYYYY_m1_2_3_modenyc/ Per-run summaries
│   │   ├── citibike_trips_by_*.csv Usage patterns
│   │   ├── axa_partner_scorecard_500m.csv EB MODEL
│   │   ├── axa_target_windows_*.csv OUTPUT
│   │   ├── latest_nyc/ SYMLINK → latest NYC
│   │   └── compare/ Multi-run comparisons
│   └── reports/ HTML reports & charts
│       ├── yYYYY_m1_2_3_modenyc/
│       │   ├── 06_insurer_story.executed.html Main report
│       │   ├── 07_risk_deep_dive.executed.html Technical analysis
│       │   ├── figures/ All charts (PNG)
│       │   └── The same for JC
│       ├── latest_nyc/ SYMLINK → latest
│       └── latest_jc/ SYMLINK → latest
├── notebooks/
│   ├── 06_insurer_story.ipynb Business analysis
│   └── 07_risk_deep_dive.ipynb Technical deep dive
├── scripts/ Pipeline automation
│   ├── download_tripdata.py Download CitiBike data
│   ├── filter_nypd_crashes.py Filter crash data
│   ├── summarize_citibike_usage.py Usage summaries
│   ├── build_axa_scorecard.py EB Risk scoring
│   ├── build_axa_target_windows.py Business recommendations
│   └── aggregate_usage_summaries.py Multi-year comparison
├── src/
│   └── ingest_tripdata.py ZIP → Parquet conversion
├── Makefile Pipeline orchestration
├── requirements.txt Python dependencies
└── README.md
```

Directory Python script CSV output Notebook HTML report Symlink EB Empirical Bayes model

Analysis

Years :

2019
pre Covid

2020
During Covid

2023 2025
post Covid

Moths : All 12 Months

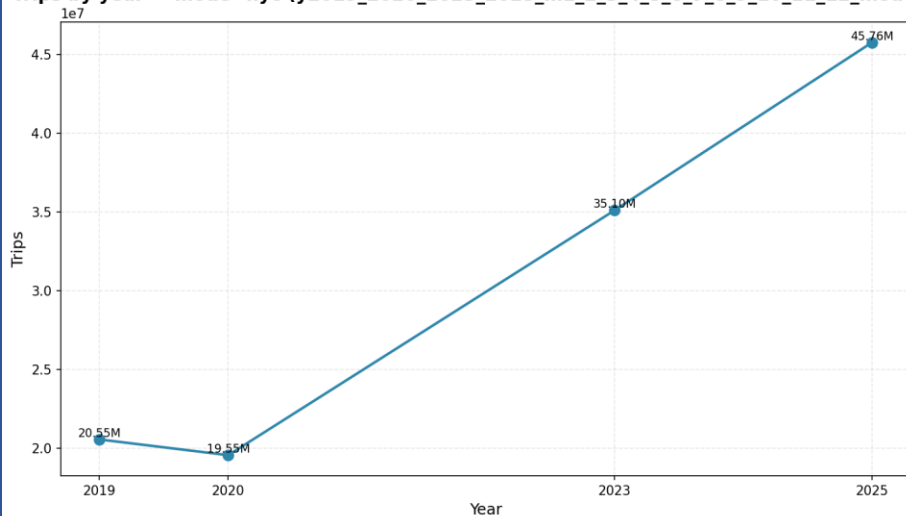
Radius: 500m and 100m (The results presented here are for 500m)

YEARLY TRENDS

Yearly Usage: Dramatic Growth Trajectory

NYC Citi Bike ridership shows remarkable expansion from 2019 to 2025. After a slight pandemic dip in 2020 (19.6M trips), the system rebounded strongly with 35.1M trips in 2023, then surged to 45.8M in 2025 a 123% increase over 2019 levels.

Trips by year — mode=nyc (y2019_2020_2023_2025_m1_2_3_4_5_6_7_8_9_10_11_12_modenyc)



NYC Mode Highlights

- Wednesday leads weekday traffic at 18.3M trips
- Peak commute hours: 5PM (11.3M) and 6PM (10.6M)
- Summer months dominate with Aug-Sept peaks
- Weekday trips represent 71% of total volume

DATA ANALYTICS REPORT

Citi Bike x NYPD: Usage Patterns + Insurer Decision Assets (NYC Analysis)

45.8M

2025 NYC Trips

Peak year showing 134.03% growth from pandemic.

5.3M

September 2025

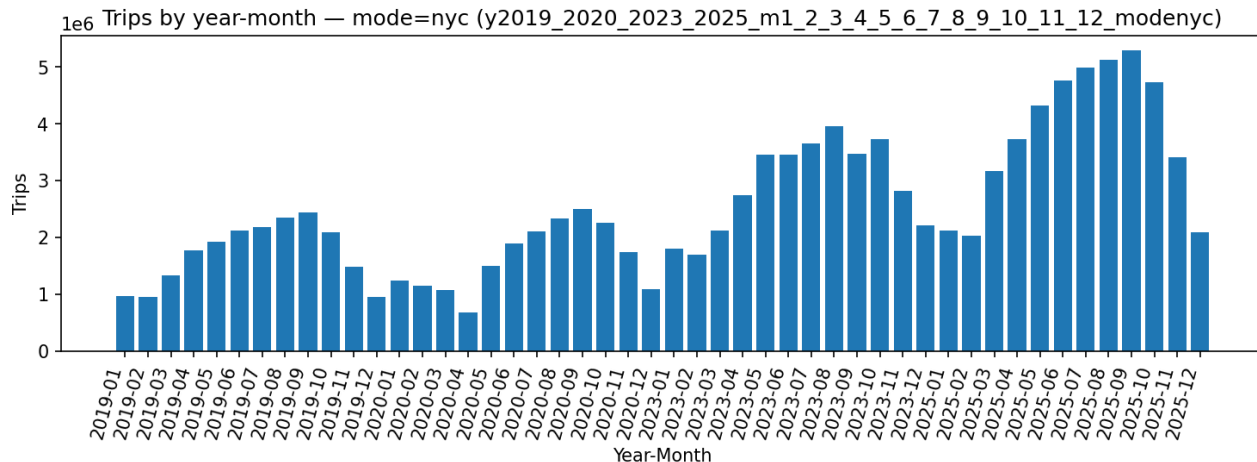
Highest monthly ridership recorded

2,669

Prevention Hotspots

High exposure + high risk stations requiring safety interventions

Monthly Usage Pattern



2019 Baseline

20.6M trips with a September peak at 2.4M. Strong summer seasonality established.

1

2023 Growth

35.1M trips showing a 70% increase. August peak reached 4.0M trips.

3

2020 Pandemic

19.6M trips with April dip to 684K. Recovery began in the summer months.

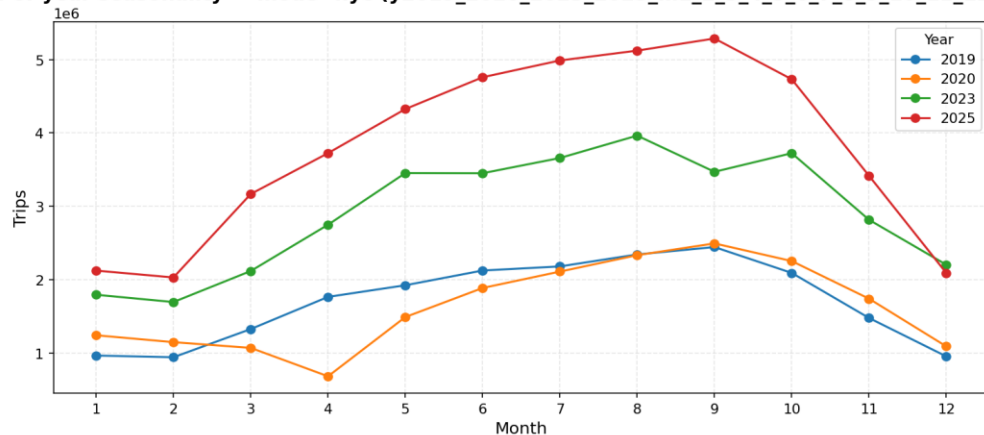
2

2025 Peak

45.8M trips with September record of 5.3M. Sustained high performance.

4

Month-of-year seasonality — mode=nyc (y2019_2020_2023_2025_m1_2_3_4_5_6_7_8_9_10_11_12_modenyc)



1

Winter Trough

Jan-Feb: 1-2M trips/month, weather-dependent ridership

2

Spring Ramp

Mar-May: 3-4M trips/month, steady growth as weather improves

3

Summer Peak

Jun-Sep: 4.5-5.3M trips/month, optimal riding conditions

4

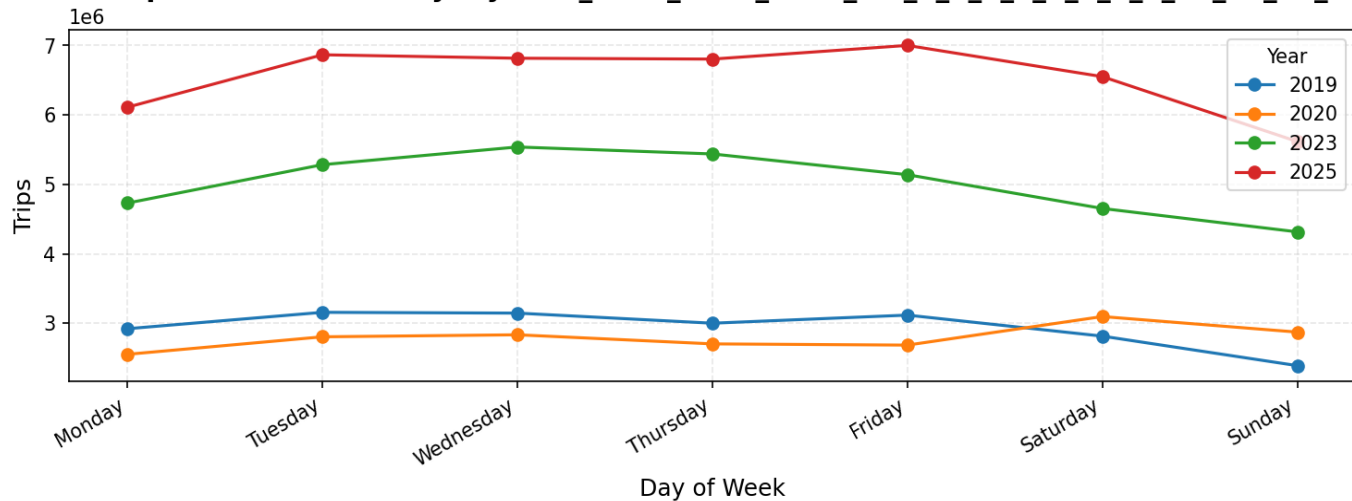
Fall Decline

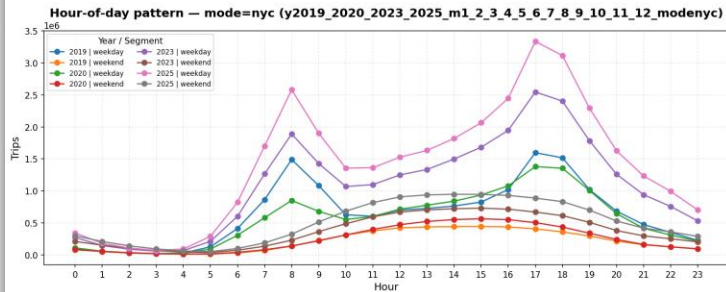
Oct-Dec: 2-4.7M trips/month, gradual seasonal decrease

Day-of-Week & Hour Patterns: Commuter-Driven System

Weekdays dominate with 71-73% of all trips across years. Tuesday and Wednesday consistently lead, each capturing ~15% of weekly volume. Weekend ridership shows distinct leisure patterns with more evenly distributed hourly usage.

Day-of-week pattern — mode=nyc (y2019_2020_2023_2025_m1_2_3_4_5_6_7_8_9_10_11_12_modenyc)





HOURLY ANALYSIS

Hour-of-Day Patterns: Actions based on partnership Ideas

1

Morning Rush

7-9am: Sharp weekday peak **Action:**

- Safety nudge:
- slow at intersections / watch turning traffic" (short, non-annoying)
- Concierge link visible but not intrusive

2

Midday Steady

10am-3pm: Moderate consistent usage, growing on weekends **Action:**

- Lighter safety reminder (especially weekends)
- Optional protection *shown softly* ("Support if something happens")

3

Evening Peak

4-7pm: Highest daily volumes, especially 5-6pm **Action:**

- Strongest safety nudges + "Had a crash?" concierge entry point prominent
- If you show the add-on here, message is support-first, not fear

4

Night Drop

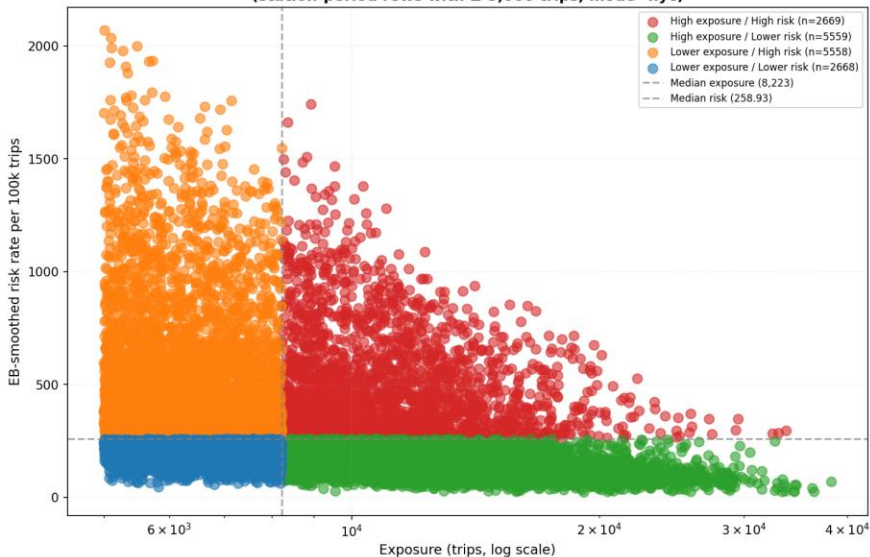
8pm-6am: Minimal usage, safety considerations increase: **Action :**

- Safety-first messaging only (visibility/road awareness)
- Concierge always available
- **Do NOT push insurance hard** (optics + lower conversion)

Crash Proximity & Risk Analysis

Empirical Bayes smoothing identifies high-risk stations by combining crash counts with exposure.

Exposure vs Risk Zones — EB-smoothed
(station-period rows with $\geq 5,000$ trips, mode=nyc)



High Exposure / High Risk

Immediate Safety Hotspots: prioritize prevention pilots, peak-hour safety nudges, and rapid incident support

Lower Exposure / High Risk

Targeted Prevention: focused safety messaging and monitoring despite lower volume early warning for hotspot escalation

High Exposure / Lower Risk

Growth Engine: best zones for opt-in protection upsell, acquisition campaigns, and scalable A/B testing.

Lower Exposure / Lower Risk

Control & Optimization: stable baseline for experiments, holdout measurement, and low-priority investment

Zone 1 anchors: High-Risk + High-Exposure hotspots repeat across years

Stations below appear in **Zone 1 (H/H)** in **2020, 2023, and 2025**, indicating persistent priority targets for safety + support interventions.

01

Forsyth St & Broome St

178,841 trips | 565.83 avg_yearly_risk rate

02

Howard St & Lafayette St

172,806 trips | 478.38 avg_yearly_risk rate

03

W 42 St & 8 Ave

284,883 trips | 477.25 avg_yearly_risk rate

04

E 45 St & 3 Ave

74,076 trips | 456.51 avg_yearly_risk rate

05

E 65 St & 2 Ave

103,430 trips | 429.91 avg_yearly_risk rate

Decision Assets: WHERE to Focus (500 m Radius)

Prevention Strategy

The locations are neighborhoods that combine high exposure with elevated crash rates, making them ideal candidates for

- **Safety Nudges (in-app):** targeted prompts during peak windows
- **Incident Concierge:** "Had a crash?" flow for guidance + optional claim initiation
- **Optional Protection (soft):** offered with *support framing* (not fear), primarily at checkout

Reach Strategy

Top exposure stations for maximum campaign impact:

1. Forsyth St & Broome St
2. Howard St & Lafayette St
3. W 42 St & 8 Ave
4. E 45 St & 3 Ave
5. E 65 St & 2 Ave

WHEN to Activate: Optimal Timing Windows

1

Seasonal Peaks

September 2025: 5.3M trips (11.6% of annual volume). August-October window captures 33% of yearly ridership.

2

Weekly Patterns

Wednesday leads at 15.2% of weekly trips. Weekdays represent 71% of total volume vs 29% weekends.

3

Daily Peaks

5PM hour: 11.3M trips (7.3% daily share). Commute hours (5-6PM)

Decision Assets: WHERE to Focus (500 m Radius VS 100 m Radius)

Prevention Strategy

The locations appeared at both stations and neighborhood levels.,

- **Safety Nudges (in-app):** targeted prompts during peak windows
- **Incident Concierge:** "Had a crash?" flow for guidance + optional claim initiation
- **Optional Protection (soft):** offered with *support framing* (not fear), primarily at checkout

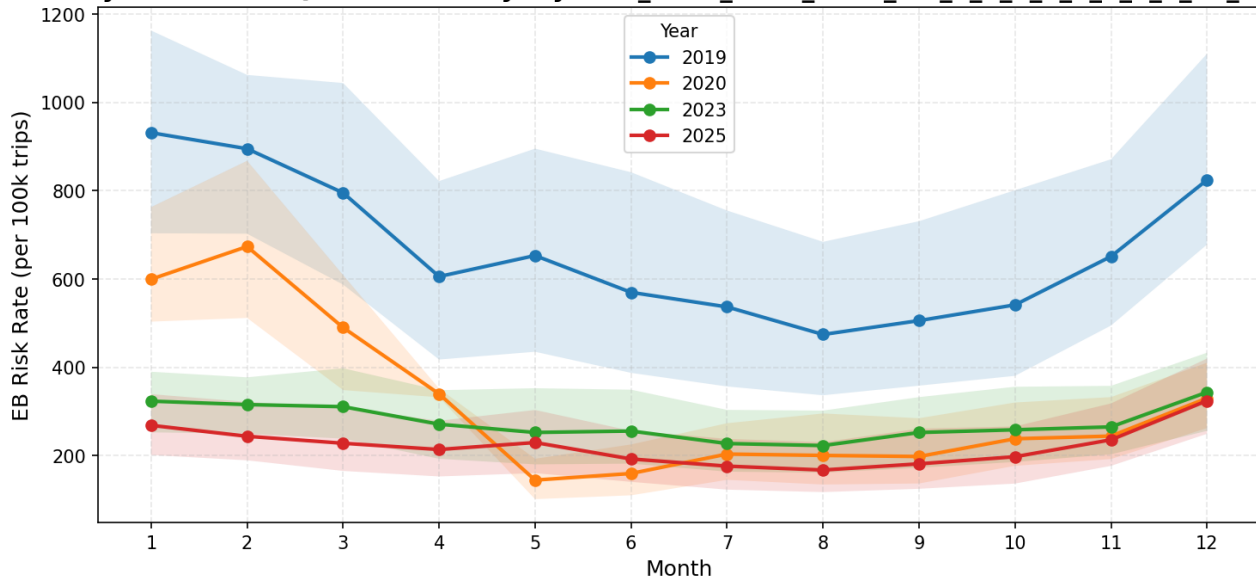
Reach Strategy

These locations need immediate action

1. Forsyth St & Broome St
2. W 42 St & 8 Ave

Risk Trajectory

Risk Seasonality (Median $\pm$ IQR) — mode=nyc (y2019_2020_2023_2025_m1_2_3_4_5_6_7_8_9_10_11_12_modenyc)



JERSEY CITY

Citi Bike JC: Parallel Growth Story

Jersey City operations mirror NYC trends with 995k trips in 2025 (up from 405k in 2019). Grove St PATH station dominates with 192k annual trips. Similar weekday commuter patterns emerge, though at smaller scale. Prevention pool includes 137 credible stations with zero expected burden due to lower crash density.

995K

2025 JC Trips

146% growth since 2019

115K

Peak Month

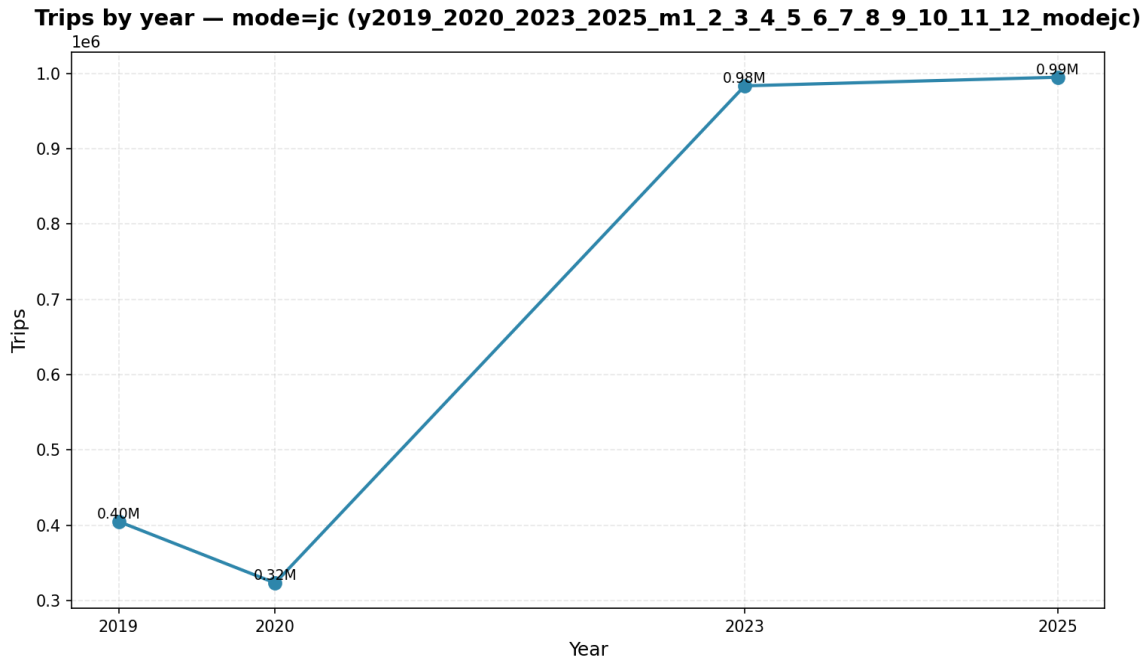
September 2025

192K

Top Station

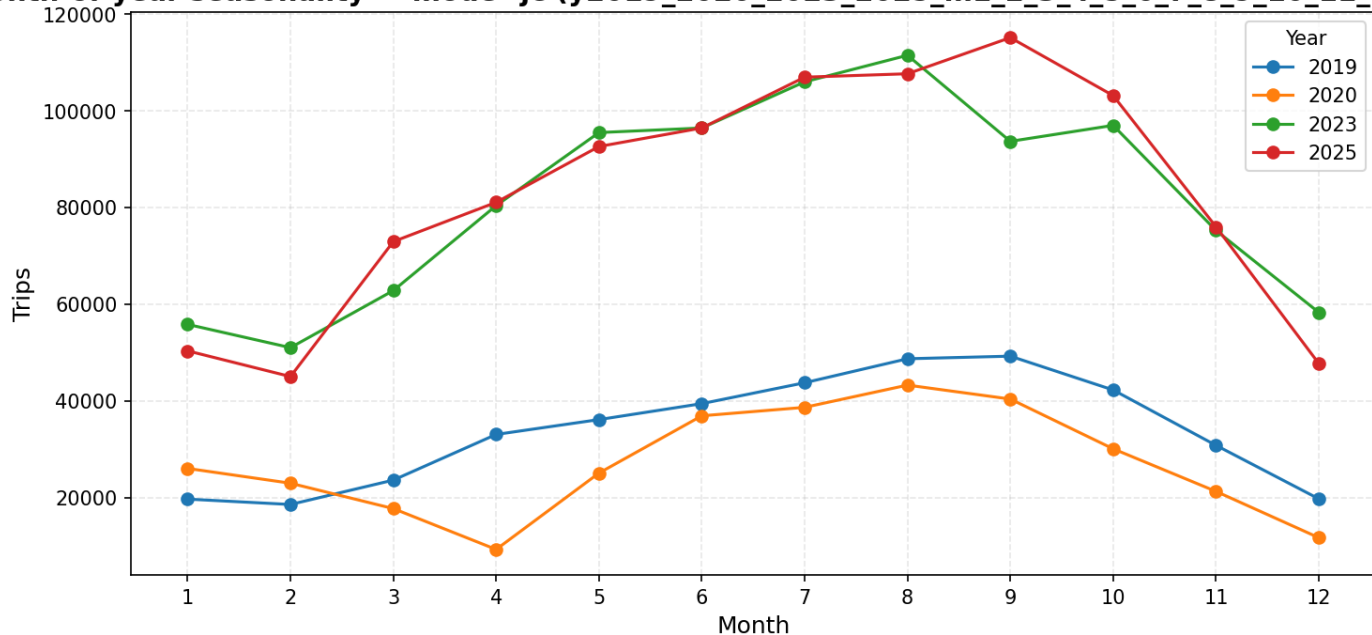
Grove St PATH

Yearly Pattern



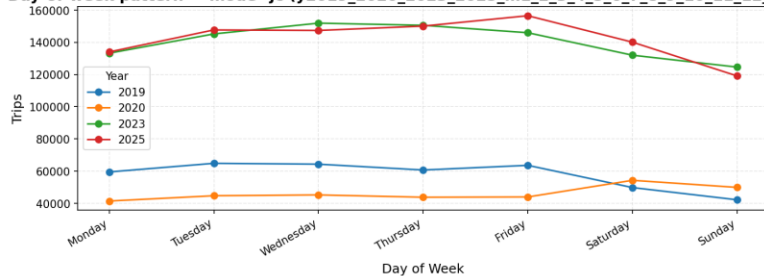
Monthly pattern

Month-of-year seasonality — mode=jc (y2019_2020_2023_2025_m1_2_3_4_5_6_7_8_9_10_11_12_modejc)

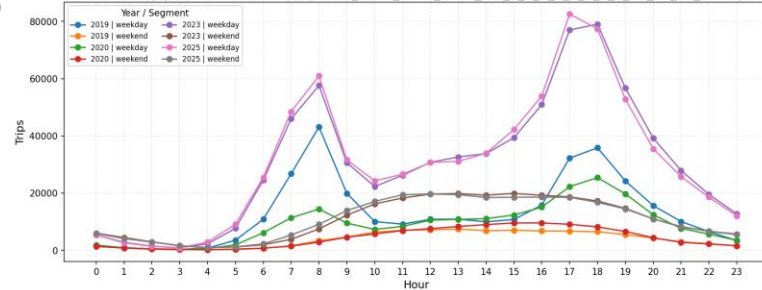


DOW and Hourly Pattern

Day-of-week pattern — mode=jc (y2019_2020_2023_2025_m1_2_3_4_5_6_7_8_9_10_11_12_modejc)



Hour-of-day pattern — mode=jc (y2019_2020_2023_2025_m1_2_3_4_5_6_7_8_9_10_11_12_modejc)



01

Day-of-Week Patterns

Friday leads with 410k trips (15.7%), strong weekday focus at 74% of total volume

02

Hourly Distribution

5-6pm evening peak with 267k and 266k trips respectively, 8am morning rush at 199k

03

Station Hierarchy

Top 20 stations range from 62k-192k annual trips, transit hubs dominate usage

Back Up

Scripts and pipeline

Citi Bike Tripdata Downloader

Reliable S3 XML listing → deterministic selection → resilient downloads

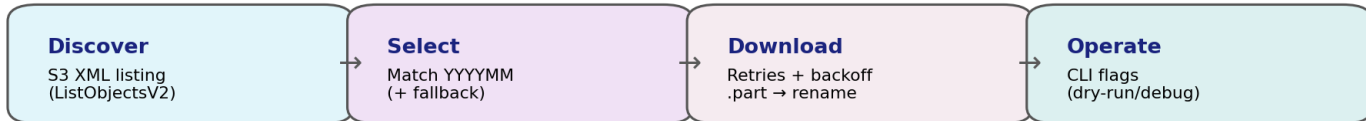
PURPOSE

Download Citi Bike tripdata ZIPs (NYC / Jersey City) reliably without scraping JS pages.

CORE MECHANICS

- ✓ **S3 ListObjectsV2 (XML) + pagination:** `_fetch_listing() → _list_bucket_zip_filenames()`
- ✓ **Filename matching by partition & mode:** NYC: YYYYMM... | JC: JC-YYYYMM... (+ fallback)
- ✓ **Robust download safety:** stream chunks + retries + `.part` → rename (atomic)
- ✓ **Operable CLI:** `--dry-run / --debug / --force / --timeout / --retries`

PIPELINE FLOW



Deterministic • Idempotent by default • Debuggable • Safe writes

Citi Bike Tripdata Ingestion (ZIP → Parquet)

Robust CSV-in-ZIP parsing → canonical schema → NYC/JC separation → atomic Parquet partitions

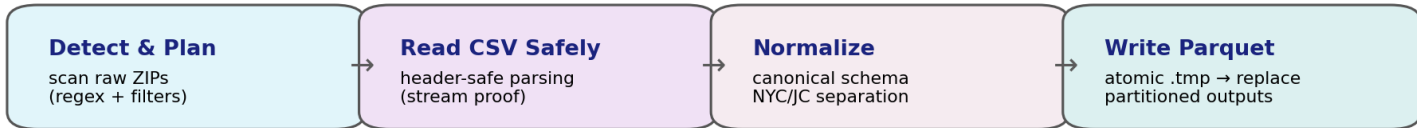
PURPOSE

Convert downloaded Citi Bike ZIP archives into clean, month-partitioned Parquet with a stable schema.

CORE MECHANICS

- ✓ **ZIP metadata detection:** regex patterns → ZipMeta; supports nested ZIPs
- ✓ **ZIP-safe CSV parsing:** read header once → rewind; pyarrow open_csv(skip_rows=1)
- ✓ **Canonicalization:** standardize names; parse timestamps; generate ride_id
- ✓ **NYC vs JC separation:** prefix filter + JC bbox fallback for legacy IDs
- ✓ **Atomic Parquet writes:** write temp .tmp → os.replace; month-based partitioning

PIPELINE FLOW



Robust to legacy formats • Deterministic partitions • Safe writes • Analysis-ready Parquet

NYPD Crash Filter (QA + Spatial-Safe Subset)

Chunked CSV → strict date/coord cleaning → year/month filter → QA report

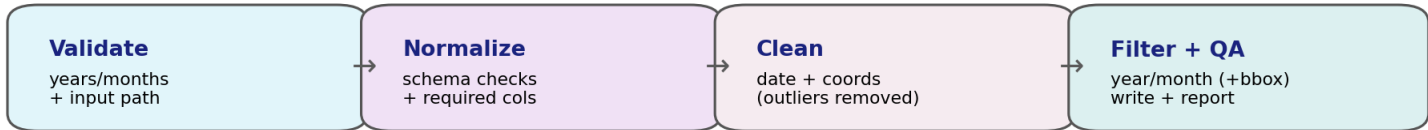
PURPOSE

Create a cleaned NYPD crash subset that is safe for station crash-proximity analysis.

CORE MECHANICS

- ✓ **Input validation:** years 2013–2030, months 1–12; avoids silent empty outputs
- ✓ **Schema defense:** normalize column names; required-column check; fast usecols
- ✓ **Spatial-safe cleaning:** drop NaN coords, drop (0,0); strict crash_date parsing
- ✓ **Filtering + NYC bbox:** subset by year/month; bbox removes distant crashes
- ✓ **QA report (auditability):** chunk-level tracking of removed rows + retention metrics

PIPELINE FLOW



Memory-safe chunks • Strict parsing • Coordinate outlier removal • QA/audit trail

Citi Bike Usage & Summary

Aggregating trip and crash data for analytics & insurance

PURPOSE

Aggregate Citi Bike trip & crash data for analytics and insurance risk scoring

ALGORITHM & FLOW

1. Read partitioned Parquet tripdata
2. Flexible schema detection (Dynamic Mapping)
3. Aggregate usage by time, station & demographic
4. Compute station exposure (Duration & Frequency)
5. Crash proximity analysis (BallTree Spatial Join)
6. Output multi-level CSVs & Risk Indices

KEY OUTPUTS

- citibike_trips_by_month.csv
- citibike_trips_by_year.csv
- citibike_trips_by_dow.csv
- citibike_trips_by_hour.csv
- citibike_station_exposure.csv
- station_risk_exposure_plus_crashproximity.csv
- summary_highlights.md

Schema Check Architecture

Validation pipeline · partition format · column resolution · crash CSV integrity

Schema Check (2)

6 validation steps · A ~ F

A

Check Partition Format (mode / year / month in path)

| regex definition + path parsing

```
# regex definition
_PART_RE = re.compile(
    r"mode=([a-zA-Z])/year=(\d{4})/"
    r"month=(\d{1,2})/"
    r"tripdata\.parquet$"
)
```

```
# path parsing
part = parse_partition(fp)
if not part:
    continue
## Skips file if path doesn't match expected pattern
```

B

Check What Columns Exist in Each Parquet File

| Used in both station + time summaries

```
cols = parquet_available_columns(fp)
🔍 Inspects file schema before reading → drives steps C and D
```

C

Choose Correct Time Column (schema variations)

| ⚡ compute_dow_and_hour()

```
time_col = _pick_time_col(cols)
if not time_col:
    continue
## Skips file if no recognised time column is found
```

D

Choose Station Columns (schema variations)

| ⚡ compute_station_exposure()

```
start_id, start_name,
start_lat, start_lng,
end_id, end_name,
end_lat, end_lng =
_pick_sta ... (cols)
```

🔍 Resolves 8 station columns across naming variations

E

Skip Files Missing Required Station Columns

| ⚡ compute_station_exposure()

```
if not start_id
or not start_name
or not end_id
or not end_name:
    continue
```

🛡️ Guards against files missing the 4 required station identifiers

F

Crash CSV Schema Check (date + lat/lng must exist)

| ⚡ _load_crashes_with_time()

```
date_col =
    first_existing_column(...)
if not date_col:
    raise RuntimeError(...)
```

```
lat_col =
    first_existing_column(...)
lng_col =
    first_existing_column(...)
if not lat_col
or not lng_col:
    raise RuntimeError(...)
🚨 Hard fail — crash data is unusable without date or coordinates
```

Station Risk Scorecard

Empirical Bayes Smoothing

PURPOSE

Rank NYC/Jersey City bike stations by insurance risk using trip exposure + crash proximity data

EMPIRICAL BAYES SHRINKAGE

$$r_EB = (Crashes + Baseline_Rate \times m) / (Trips + m)$$

Where m = "prior strength" (auto-calibrated, bounded [1k-100k])

WHY EMPIRICAL BAYES?

- ✓ Prevents outliers from distorting rankings (1 crash @ 50 trips $\neq$ 50 crashes @ 100k trips)
- ✓ Shrinks risky estimates toward group baseline (Poisson regularization)
- ✓ Transparent outputs: Both raw CI and smoothed EB estimates provided

FIVE-LAYER OUTLIER DEFENSE

Credibility

Trips $\geq$ min_trips

EB Shrinkage

Poisson rate smoothing

Signal Detection

Variance test $\rightarrow$ fallback

Bounded Prior

$m \in [1k, 100k]$

Mode Isolation

NYC/JC separate

KEY OUTPUTS

- **axa_priority_score:** 0-1 percentile
- **credibility_flag:** credible/insufficient
- **prevention_hotspot:** High risk + high volume

Data Flow: Input → Processing → Output

INPUTS

station_risk_exposure_plus_crashproximity.csv

- └ Station IDs & names
- └ Lat/Lng coordinates
- └ Crashes within radius (500m, 750m, 1km...)
- └ Trips by station
- └ Raw crash rates per 100k trips

PROCESSING STEPS

- ☐ **1. Validate** Check required columns, coerce types
- ☐ **2. Poisson CI** Compute exact confidence intervals on raw rates
- ☐ **3. EB Prior** Auto-calibrate m using exposure-weighted variance
- ☐ **4. EB Smoothing** Apply shrinkage formula grouped by mode/year/month
- ☐ **5. Ranking** Percentile-rank risk & exposure within groups
- ☐ **6. Hotspot Detection** Flag prevention/product/acquisition hotspots

- Scorecard

OUTPUTS

axa_partner_scorecard_<radius>m.csv

- └ 26 columns including:
- └ axa_priority_score, exposure_pct, risk_pct
- └ credibility_flag, risk_proxy_available
- └ eb_risk_rate_per_100k_trips
- └ prevention/product/acquisition_hotspot
- └ Sorted by priority_score desc.

Design Decisions & Edge Cases

KEY DESIGN DECISIONS

► Auto-calibrate m	Rather than fixed value	<i>(Adapts to data variance)</i>
► Mode isolation	Process NYC/JC separately	<i>(Prevents regional confounds)</i>
► min_trips = 5k	Credibility threshold	<i>(Balance: confidence vs. coverage)</i>
► Bounded prior	$m \in [1k, 100k]$	<i>(Prevent extreme shrinkage)</i>
► Exposure-weighted EB	Prior calibration method	<i>(High-volume stations drive estimate)</i>

- scorecard

EDGE CASES HANDLED ✓

- ✓ Stations with 0 crashes → Works (EB shrinks to baseline)
- ✓ Mix of credible/insufficient stations → Branching logic isolates ranking
- ✓ No risk signal detected → Fallback to exposure_only mode
- ✓ Small groups ($n < 30$) → Strong prior fallback ($m = 20,000$)

This script demonstrates rigorous statistical design + practical engineering

AXA Target Windows Builder (When to Act)

Trip summaries → aggregate across years → pick top windows → action-ready table

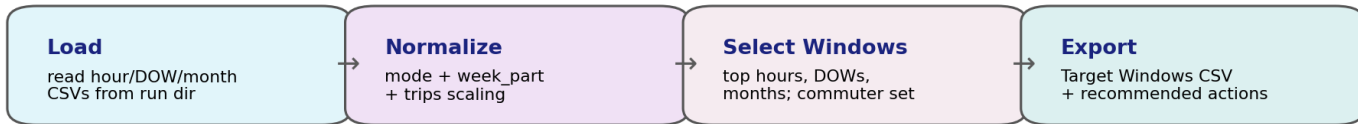
PURPOSE

Transform run-level time summaries into a unified “when to act” table for targeting & operations.

CORE MECHANICS

- ✓ **Input normalization:** requires 3 summary CSVs; enforces single mode (NYC/JC)
- ✓ **Year-robust targeting:** aggregates trips across years for stable, non-specific windows
- ✓ **Priority metrics:** ranks hour/DOW by within-week-part share (weekday vs weekend)
- ✓ **Window selection:** identifies top hours, commuter sets, and high-volume months
- ✓ **Action-ready output:** exports axa_target_windows.csv with recommended_action labels

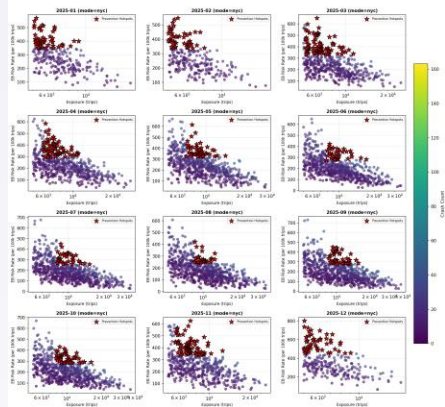
PIPELINE FLOW



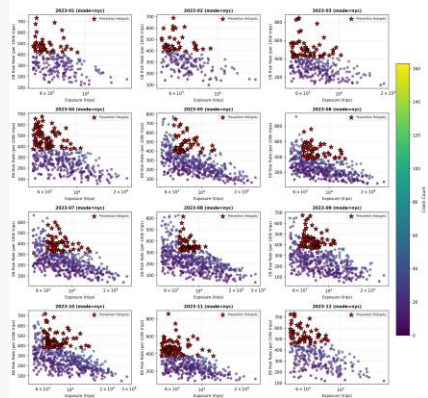
Year-robust windows • Segment-aware percentages • Clear prioritization • Campaign-ready actions

Extra Plots and Tables for NYC

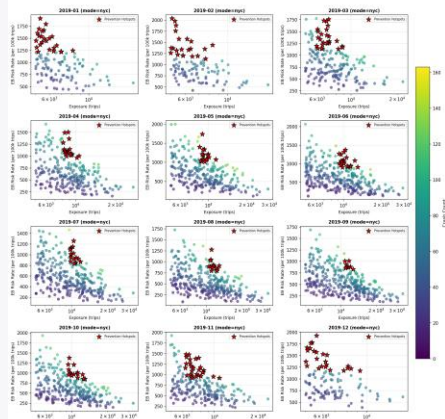
Risk vs Exposure – mode-nyrc (page 4/4)



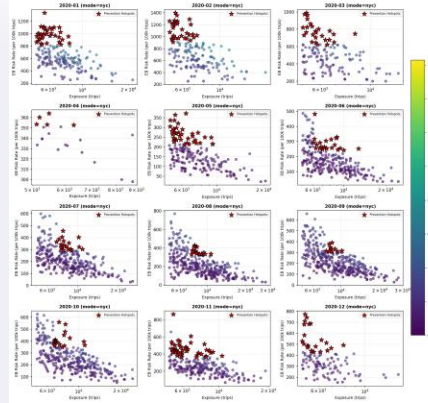
Risk vs Exposure – mode-nyrc (page 5/4)



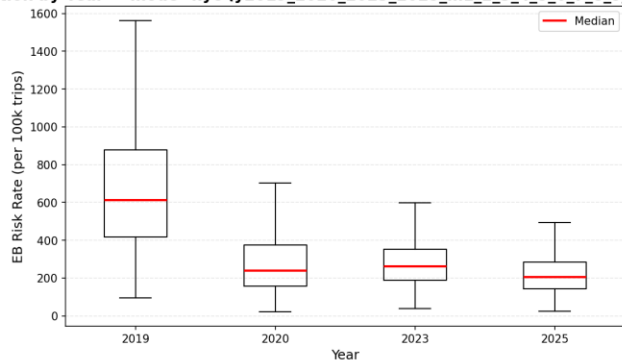
Risk vs Exposure – mode-nyrc (page 1/4)



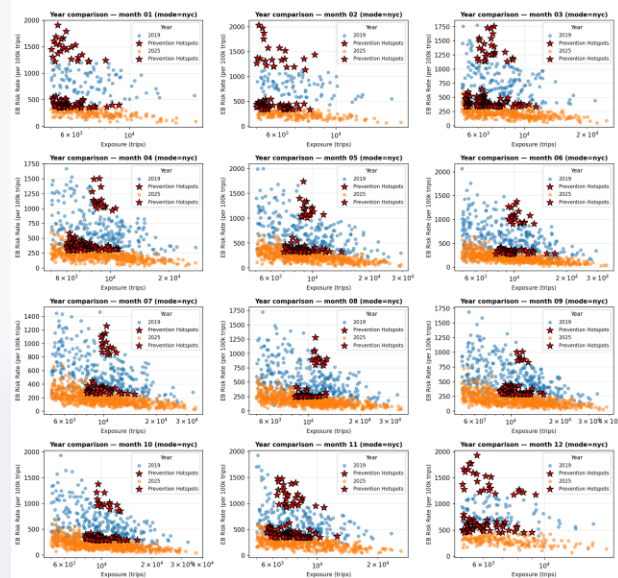
Risk vs Exposure – mode-nyrc (page 2/4)



Risk Distribution by Year — mode=nyc (y2019_2020_2023_2025_m1_2_3_4_5_6_7_8_9_10_11_12_modenyc)



Yearly comparison (overlay: endpoints_2019_2025) — mode=nyc



Top 20 by EB exposure:						
	mode	year	month	station_id	station_name	trips
1638	nyc	2019	9	519	Pershing Square North	33714
2394	nyc	2019	8	519	Pershing Square North	33019
4942	nyc	2019	10	519	Pershing Square North	32695
2870	nyc	2019	7	519	Pershing Square North	32472
29082	nyc	2019	9	426	West St & Chambers St	30616
12488	nyc	2019	9	285	Broadway & E 14 St	30488
9078	nyc	2019	9	497	E 17 St & Broadway	29639
2122	nyc	2019	6	519	Pershing Square North	29474
5578	nyc	2019	5	519	Pershing Square North	29320
13697	nyc	2019	8	497	E 17 St & Broadway	29249
38395	nyc	2019	8	426	West St & Chambers St	28605
22631	nyc	2019	6	426	West St & Chambers St	27573
30470	nyc	2019	8	514	12 Ave & W 40 St	27453
6120	nyc	2019	9	3255	8 Ave & W 31 St	27406
2515	nyc	2019	4	519	Pershing Square North	27173
18435	nyc	2019	6	514	12 Ave & W 40 St	27126
29352	nyc	2019	9	514	12 Ave & W 40 St	26671
34974	nyc	2019	7	426	West St & Chambers St	26514
10517	nyc	2019	9	435	W 21 St & 6 Ave	26434
23521	nyc	2019	9	358	Christopher St & Greenwich St	26249

--- Year 2020 --- Top 20 by exposure:						
	mode	year	month	station_id	station_name	trips
58548	nyc	2020	8	6765.01	12 Ave & W 40 St	28799
22757	nyc	2020	9	6140.05	W 21 St & 6 Ave	27859
58817	nyc	2020	8	5329.03	West St & Chambers St	27151
44907	nyc	2020	9	5329.03	West St & Chambers St	27048
63774	nyc	2020	9	6765.01	12 Ave & W 40 St	25811
546	nyc	2020	9	6822.09	1 Ave & E 68 St	25537
57391	nyc	2020	7	6765.01	12 Ave & W 40 St	25498
50706	nyc	2020	6	6765.01	12 Ave & W 40 St	24632
23446	nyc	2020	8	6140.05	W 21 St & 6 Ave	24620
63213	nyc	2020	7	5329.03	West St & Chambers St	24530
19668	nyc	2020	10	6140.05	W 21 St & 6 Ave	24192
271	nyc	2020	10	6822.09	1 Ave & E 68 St	24104
1260	nyc	2020	8	6822.09	1 Ave & E 68 St	23248
54764	nyc	2020	8	5696.03	Pier 40 - Hudson River Park	23115
29487	nyc	2020	8	5847.01	Christopher St & Greenwich St	22733
23286	nyc	2020	9	5980.07	E 17 St & Broadway	22728
3332	nyc	2020	7	6822.09	1 Ave & E 68 St	22271
20595	nyc	2020	9	6948.11	Broadway & W 60 St	22158
51078	nyc	2020	6	5329.03	West St & Chambers St	22087
838	nyc	2020	6	6822.09	1 Ave & E 68 St	21913

Top 20 by EB risk rate:								
	mode	year	month	station_id	station_name	trips	crash_count	eb_risk_rate_per_100k_trips
1938	nyc	2019	6	3734	E 58 St & 1 Ave (NW Corner)	5008	116	2068.172197
308	nyc	2019	2	529	W 42 St & 8 Ave	5098	109	2035.034813
1843	nyc	2019	5	3712	W 35 St & Dyer Ave	5486	120	1999.473641
3297	nyc	2019	5	3749	Lexington Ave & E 36 St	5108	112	1992.237399
213	nyc	2019	2	401	Allen St & Rivington St	5270	109	1979.209296
802	nyc	2019	10	464	E 56 St & 3 Ave	5731	122	1934.592498
43	nyc	2019	12	529	W 42 St & 8 Ave	5672	112	1931.177105
609	nyc	2019	11	3236	W 42 St & Dyer Ave	5081	106	1926.525165
496	nyc	2019	1	529	W 42 St & 8 Ave	5319	105	1907.926813
690	nyc	2019	2	531	Forsyth St & Broome St	5339	104	1878.788813
307	nyc	2019	1	167	E 39 St & 3 Ave	5722	105	1793.542031
602	nyc	2019	12	2021	W 45 St & 8 Ave	5396	97	1779.989626
964	nyc	2019	3	305	E 58 St & 3 Ave	5930	111	1776.686078
684	nyc	2019	12	167	E 39 St & 3 Ave	5313	95	1771.711333
3504	nyc	2019	3	3466	W 45 St & 6 Ave	5087	95	1759.887387
2759	nyc	2019	6	3749	Lexington Ave & E 36 St	5607	108	1759.585071
38	nyc	2019	3	529	W 42 St & 8 Ave	7140	131	1758.284339
1254	nyc	2019	2	167	E 39 St & 3 Ave	5321	96	1757.576695
27	nyc	2019	5	529	W 42 St & 8 Ave	8920	163	1740.784882
495	nyc	2019	3	448	W 37 St & 10 Ave	6629	120	1731.869776

Top 20 by EB risk rate:								
	mode	year	month	station_id	station_name	trips	crash_count	eb_risk_rate_per_100k_trips
583	nyc	2020	2	6753.08	1 Ave & E 62 St	5423	78	1399.579388
660	nyc	2020	2	5523.02	Stanton St & Chrystie St	5533	77	1360.706936
423	nyc	2020	1	6762.02	E 58 St & 3 Ave	5651	78	1336.319570
177	nyc	2020	2	6602.05	W 42 St & 8 Ave	6579	86	1291.660959
1698	nyc	2020	2	6650.07	E 55 St & 2 Ave	5367	69	1270.535324
3060	nyc	2020	2	6593.13	W 45 St & 6 Ave	5018	64	1261.133003
1381	nyc	2020	2	5453.05	Forsyth St & Broome St	5587	71	1258.463399
2656	nyc	2020	2	5391.06	Lispenard St & Broadway	5317	66	1233.100904
2738	nyc	2020	2	6441.05	Broadway & W 38 St	5277	65	1225.027626
1779	nyc	2020	2	5414.06	Allen St & Rivington St	5573	68	1215.502573
3372	nyc	2020	2	6551.02	W 44 St & 5 Ave	5201	63	1207.788810
263	nyc	2020	2	6389.03	E 39 St & 3 Ave	6697	81	1206.898585
3292	nyc	2020	2	5561.04	Mott St & Prince St	5299	63	1188.998001
3217	nyc	2020	2	6611.02	W 37 St & 10 Ave	5306	63	1187.678150
4331	nyc	2020	2	6551.03	E 43 St & 5 Ave	5259	60	1148.665667
2339	nyc	2020	2	6617.09	E 53 St & Lexington Ave	5940	66	1122.406111
2712	nyc	2020	1	5422.04	Howard St & Lafayette St	5350	60	1116.198655
2075	nyc	2020	1	5523.02	Stanton St & Chrystie St	5559	62	1111.123869
575	nyc	2020	1	6753.08	1 Ave & E 62 St	6513	72	1103.136092
175	nyc	2020	1	6602.05	W 42 St & 8 Ave	7062	78	1102.438782

Top 26 by exposure:						
	mode	year	month	station_id	station_name	trips
8703	nyc	2023	10	6140.05	W 21 St & 6 Ave	28909
2292	nyc	2023	8	6140.05	W 21 St & 6 Ave	28571
42013	nyc	2023	8	5329.03	West St & Chambers St	28520
5595	nyc	2023	7	6140.05	W 21 St & 6 Ave	26824
3506	nyc	2023	9	6140.05	W 21 St & 6 Ave	26739
7652	nyc	2023	8	6726.01	11 Ave & W 41 St	26425
16603	nyc	2023	6	6140.05	W 21 St & 6 Ave	26399
34927	nyc	2023	10	5329.03	West St & Chambers St	26122
4841	nyc	2023	5	6140.05	W 21 St & 6 Ave	25394
2610	nyc	2023	11	6140.05	W 21 St & 6 Ave	25373
45630	nyc	2023	7	5329.03	West St & Chambers St	25270
29820	nyc	2023	9	5329.03	West St & Chambers St	24587
9884	nyc	2023	8	5980.07	E 17 St & Broadway	23958
34095	nyc	2023	5	5329.03	West St & Chambers St	23761
34842	nyc	2023	6	5329.03	West St & Chambers St	23521
6775	nyc	2023	10	5905.14	University Pl & E 14 St	23305
8254	nyc	2023	7	6726.01	11 Ave & W 41 St	23146
11178	nyc	2023	8	6912.01	7 Ave & Central Park South	23107
19092	nyc	2023	8	6459.07	W 30 St & 10 Ave	22808
6662	nyc	2023	7	6912.01	7 Ave & Central Park South	22725

Top 26 by exposure:						
	mode	year	month	station_id	station_name	trips
5427	nyc	2025	9	6140.05	W 21 St & 6 Ave	38294
38601	nyc	2025	9	5329.03	West St & Chambers St	36460
21608	nyc	2025	10	6140.05	W 21 St & 6 Ave	36315
12684	nyc	2025	8	5788.13	Lafayette St & E 8 St	34690
13317	nyc	2025	9	5788.13	Lafayette St & E 8 St	34609
41662	nyc	2025	8	5329.03	West St & Chambers St	34523
38663	nyc	2025	9	6233.04	Pier 61 at Chelsea Piers	34490
22253	nyc	2025	6	5329.03	West St & Chambers St	33726
37646	nyc	2025	8	6233.04	Pier 61 at Chelsea Piers	33676
11262	nyc	2025	7	6140.05	W 21 St & 6 Ave	33596
2332	nyc	2025	8	6140.05	W 21 St & 6 Ave	33395
42466	nyc	2025	7	6233.04	Pier 61 at Chelsea Piers	33372
25436	nyc	2025	6	6140.05	W 21 St & 6 Ave	32918
10995	nyc	2025	9	6876.04	Central Park 5 & 6 Ave	32917
38690	nyc	2025	7	5329.03	West St & Chambers St	32669
6997	nyc	2025	10	5788.13	Lafayette St & E 8 St	31733
1299	nyc	2025	9	6492.08	9 Ave & W 33 St	31515
36800	nyc	2025	6	6233.04	Pier 61 at Chelsea Piers	31496
13480	nyc	2025	9	5905.14	University Pl & E 14 St	31128
5099	nyc	2025	8	6912.01	7 Ave & Central Park South	30675

Top 26 by EB risk rate								
	mode	year	month	station_id	station_name	trips	crash_count	eb_risk_rate_per_100k_trips
290	nyc	2023	6	6560.14	W 40 St & 7 Ave	5580	56	923.541404
134	nyc	2023	11	5453.05	Forsyth St & Broome St	5631	51	858.123905
172	nyc	2023	3	5343.11	Orchard St & Grand St	5673	49	845.528915
294	nyc	2023	3	5445.07	Stanton St & Norfolk St	5635	48	835.299842
406	nyc	2023	3	5523.02	Stanton St & Chrystie St	5463	46	826.584319
37	nyc	2023	3	5414.06	Allen St & Rivington St	7349	58	783.592580
405	nyc	2023	5	5351.07	Walker St & Baxter St	6121	50	773.561520
97	nyc	2023	6	5453.05	Forsyth St & Broome St	7249	58	760.928893
475	nyc	2023	5	6762.04	3 Ave & E 62 St	6180	49	753.277380
1047	nyc	2023	5	6569.07	W 36 St & 9 Ave	5277	42	750.124516
163	nyc	2023	11	5453.01	Rivington St & Chrystie St	6348	49	747.171967
364	nyc	2023	5	5453.05	Forsyth St & Broome St	6393	50	745.100986
1236	nyc	2023	5	5553.10	E 2 St & Avenue A	5195	41	743.911475
232	nyc	2023	1	5406.02	Rivington St & Ridge St	5246	38	732.346164
686	nyc	2023	12	5453.01	Rivington St & Chrystie St	5293	38	729.529946
552	nyc	2023	6	5453.06	Kenmare St & Elizabeth St	6399	49	726.706641
1143	nyc	2023	12	6762.04	3 Ave & E 62 St	5054	36	725.294343
649	nyc	2023	12	5523.02	Stanton St & Chrystie St	5334	38	724.807697
527	nyc	2023	10	4479.06	Bond St & Fulton St	6281	47	719.067108
1692	nyc	2023	5	5342.10	Allen St & Hester St	5155	39	716.252086

Top 26 by EB risk rate:								
	mode	year	month	station_id	station_name	trips	crash_count	eb_risk_rate_per_100k_trips
376	nyc	2025	12	6398.06	W 37 St & 5 Ave	5086	41	801.824669
903	nyc	2025	12	6560.01	Broadway & W 41 St	5073	37	737.675768
875	nyc	2025	9	6441.06	W 37 St & Broadway	5471	44	730.374131
676	nyc	2025	12	5523.02	Stanton St & Chrystie St	5314	38	725.357133
1364	nyc	2025	9	4668.08	Johnson St & Gold St	5130	41	722.063785
639	nyc	2025	12	6626.01	E 48 St & 5 Ave	5518	38	702.654946
281	nyc	2025	12	6560.15	Broadway & W 38 St	6158	42	695.711782
1067	nyc	2025	12	6593.15	6 Ave & W 45 St	5374	36	687.151700
57	nyc	2025	7	5414.07	Delancey St & Eldridge St	5838	43	676.803992
912	nyc	2025	10	5414.07	Delancey St & Eldridge St	5355	39	669.408058
895	nyc	2025	5	6569.07	W 36 St & 9 Ave	5458	39	668.434634
746	nyc	2025	7	4668.08	Johnson St & Gold St	5198	38	666.018989
185	nyc	2025	12	6602.05	W 42 St & 8 Ave	6706	43	659.214240
53	nyc	2025	12	6708.04	Broadway & W 48 St	7187	46	657.127756
1164	nyc	2025	12	4637.06	Fulton St & Adams St	5548	35	653.620180
1513	nyc	2025	5	6569.08	W 35 St & Dyer Ave	5183	36	649.644326
77	nyc	2025	3	6644.07	W 42 St & Dyer Ave	6236	42	649.274227
580	nyc	2025	6	5414.07	Delancey St & Eldridge St	5870	41	648.669290
442	nyc	2025	5	4437.01	3 Ave & Schermerhorn St	6320	43	644.364873
1650	nyc	2025	12	6551.02	W 44 St & 5 Ave	5361	33	641.393639

Extra Tables for JC

Top 20 by exposure:						
	mode	year	month	station_id	station_name	trips
	0	jc	2019	8	3186	Grove St PATH 12143
	1	jc	2019	9	3186	Grove St PATH 11830
	2	jc	2019	10	3186	Grove St PATH 11453
	3	jc	2019	7	3186	Grove St PATH 11104
	9	jc	2019	6	3186	Grove St PATH 9728
	14	jc	2019	5	3186	Grove St PATH 9398
	17	jc	2019	11	3186	Grove St PATH 8639
	20	jc	2019	4	3186	Grove St PATH 8305
	27	jc	2019	3	3186	Grove St PATH 6599
	73	jc	2019	9	3203	Hamilton Park 5966
	34	jc	2019	1	3186	Grove St PATH 5647
	74	jc	2019	8	3203	Hamilton Park 5555
	37	jc	2019	12	3186	Grove St PATH 5289
	38	jc	2019	2	3186	Grove St PATH 5221
	76	jc	2019	7	3203	Hamilton Park 5122
	77	jc	2019	10	3203	Hamilton Park 5073
	145	jc	2019	9	3792	Columbus Dr at Exchange Pl 4854
	146	jc	2019	8	3792	Columbus Dr at Exchange Pl 4807
	215	jc	2019	8	3639	Harborside 4696
	88	jc	2019	6	3203	Hamilton Park 4653

Top 20 by exposure:						
	mode	year	month	station_id	station_name	trips
	4	jc	2023	8	JC115	Grove St PATH 10987
	5	jc	2023	7	JC115	Grove St PATH 10680
	7	jc	2023	6	JC115	Grove St PATH 10202
	10	jc	2023	10	JC115	Grove St PATH 9700
	12	jc	2023	5	JC115	Grove St PATH 9527
	58	jc	2023	10	HB102	Hoboken Terminal - River St & Hudson Pl 8901
	65	jc	2023	7	HB102	Hoboken Terminal - River St & Hudson Pl 8595
	19	jc	2023	9	JC115	Grove St PATH 8535
	66	jc	2023	5	HB102	Hoboken Terminal - River St & Hudson Pl 8183
	59	jc	2023	8	HB102	Hoboken Terminal - River St & Hudson Pl 7935
	53	jc	2023	9	HB102	Hoboken Terminal - River St & Hudson Pl 7890
	113	jc	2023	7	HB103	South Waterfront Walkway - Sinatra Dr & 1 St 7836
	67	jc	2023	6	HB102	Hoboken Terminal - River St & Hudson Pl 7785
	106	jc	2023	8	HB103	South Waterfront Walkway - Sinatra Dr & 1 St 7721
	22	jc	2023	4	JC115	Grove St PATH 7638
	23	jc	2023	11	JC115	Grove St PATH 7490
	61	jc	2023	11	HB102	Hoboken Terminal - River St & Hudson Pl 7317
	68	jc	2023	4	HB102	Hoboken Terminal - River St & Hudson Pl 7006
	114	jc	2023	6	HB103	South Waterfront Walkway - Sinatra Dr & 1 St 6950
	115	jc	2023	5	HB103	South Waterfront Walkway - Sinatra Dr & 1 St 6812

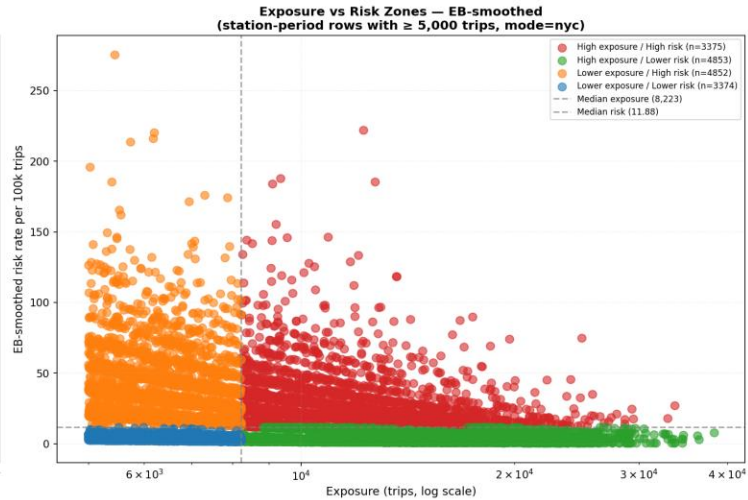
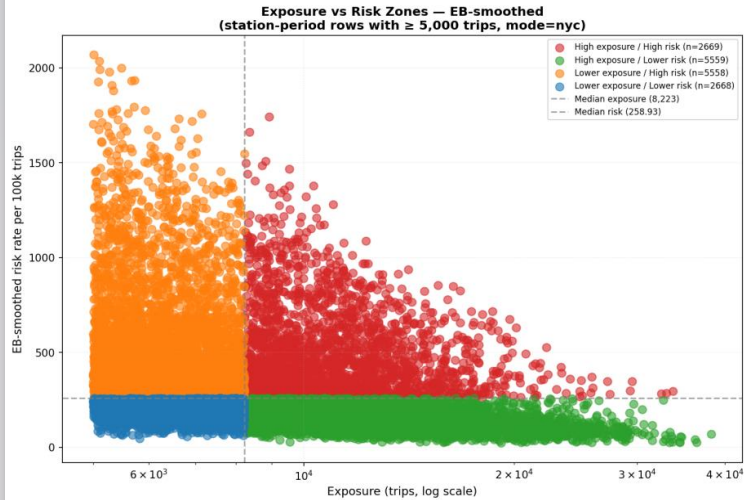
Top 20 by exposure:						
	mode	year	month	station_id	station_name	trips
	26	jc	2020	1	3186	Grove St PATH 6648
	30	jc	2020	8	3199	Newport Pkwy 6100
	33	jc	2020	9	3186	Grove St PATH 5721
	35	jc	2020	2	3186	Grove St PATH 5437
	86	jc	2020	8	3186	Grove St PATH 5416
	36	jc	2020	7	3199	Newport Pkwy 5373
	39	jc	2020	6	3199	Newport Pkwy 5221
	87	jc	2020	6	3192	Liberty Light Rail 5210
	75	jc	2020	9	3199	Newport Pkwy 5158
	158	jc	2020	8	3192	Liberty Light Rail 4823
	42	jc	2020	5	3192	Liberty Light Rail 4795
	78	jc	2020	7	3192	Liberty Light Rail 4668
	147	jc	2020	7	3186	Grove St PATH 4658
	43	jc	2020	10	3186	Grove St PATH 4498
	148	jc	2020	9	3192	Liberty Light Rail 4125
	159	jc	2020	6	3186	Grove St PATH 3890
	230	jc	2020	8	3276	Marin Light Rail 3821
	81	jc	2020	10	3199	Newport Pkwy 3666
	288	jc	2020	8	3203	Hamilton Park 3553
	219	jc	2020	7	3203	Hamilton Park 3537

Top 20 by exposure:						
	mode	year	month	station_id	station_name	trips
	6	jc	2025	9	JC115	Grove St PATH 10246
	8	jc	2025	8	JC115	Grove St PATH 10164
	11	jc	2025	7	JC115	Grove St PATH 9577
	13	jc	2025	10	JC115	Grove St PATH 9467
	15	jc	2025	5	HB102	Hoboken Terminal - River St & Hudson Pl 9240
	16	jc	2025	6	JC115	Grove St PATH 8777
	18	jc	2025	4	HB102	Hoboken Terminal - River St & Hudson Pl 8544
	49	jc	2025	9	HB106	River St & Newark St 8235
	56	jc	2025	5	JC115	Grove St PATH 8111
	21	jc	2025	3	HB102	Hoboken Terminal - River St & Hudson Pl 8071
	60	jc	2025	4	JC115	Grove St PATH 7414
	50	jc	2025	10	HB106	River St & Newark St 7265
	24	jc	2025	11	JC115	Grove St PATH 7047
	62	jc	2025	3	JC115	Grove St PATH 6847
	52	jc	2025	8	HB106	River St & Newark St 6401
	54	jc	2025	6	HB101	Hoboken Terminal - Hudson St & Hudson Pl 6280
	97	jc	2025	9	HB101	Hoboken Terminal - Hudson St & Hudson Pl 6140
	57	jc	2025	7	HB101	Hoboken Terminal - Hudson St & Hudson Pl 6075
	31	jc	2025	1	HB102	Hoboken Terminal - River St & Hudson Pl 6074
	101	jc	2025	6	HB106	River St & Newark St 6056

$R=100\text{ m}$ vs $R=500\text{ m}$

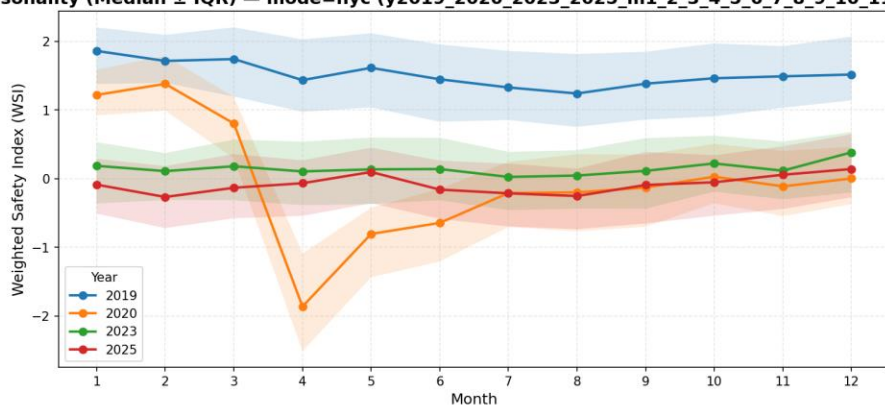
station_name	n_years_high_risk	years	avg_yearly_risk	total_trips
Forsyth St & Broome St	3	2020, 2023, 2025	565.83	178841
Howard St & Lafayette St	3	2020, 2023, 2025	478.38	172806
W 42 St & 8 Ave	3	2020, 2023, 2025	477.25	284883
E 45 St & 3 Ave	3	2020, 2023, 2025	465.51	74076
E 65 St & 2 Ave	3	2020, 2023, 2025	429.91	103430
W 37 St & Broadway	2	2023, 2025	588.08	33853
W 40 St & 7 Ave	2	2023, 2025	571.01	98595
Rivington St & Chrystie St	2	2020, 2023	544.51	110683
W 47 St & 6 Ave	2	2023, 2025	536.49	68690
St Nicholas Ave & Manhattan Ave	2	2020, 2023	526.31	61459
Stanton St & Chrystie St	2	2020, 2023	521.38	150404
3 Ave & E 62 St	2	2020, 2023	513.08	94602
E 58 St & 1 Ave (NE Corner)	2	2023, 2025	511.11	45560
W 36 St & 7 Ave	2	2023, 2025	510.14	98388
Allen St & Rivington St	2	2020, 2023	498.12	208243
Roebing St & N 4 St	2	2020, 2023	494.73	50043
E 102 St & 1 Ave	2	2020, 2023	493.34	57103
E 58 St & 1 Ave (NW Corner)	2	2020, 2023	485.94	45860
Duffield St & Willoughby St	2	2023, 2025	485.91	92577
Allen St & Hester St	2	2020, 2023	485.46	139644
W 36 St & 9 Ave	2	2023, 2025	483.61	95215
Adam Clayton Powell Blvd & W 126 St	2	2023, 2025	483.26	76134
DeKalb Ave & Hudson Ave	2	2023, 2025	478.29	76320
Bergen St & 4 Ave	2	2023, 2025	472.95	48213
W 42 St & Dyer Ave	2	2023, 2025	459.28	151486
Lawrence St & Willoughby St	2	2023, 2025	440.22	139032
W 43 St & 6 Ave	2	2023, 2025	436.57	65890
Sands St & Jay St	2	2023, 2025	419.91	77329

station_name	n_years_high_risk	years	avg_yearly_risk	total_trips	n
Atlantic Ave & Furman St	3	2020, 2023, 2025	66.10	186777	
Frederick Douglass Blvd & W 115 St	3	2020, 2023, 2025	40.39	89818	
E 97 St & Madison Ave	3	2020, 2023, 2025	34.31	140744	
W 45 St & 8 Ave	3	2020, 2023, 2025	28.24	332994	
Metropolitan Ave & Meeker Ave	2	2023, 2025	60.44	128974	
Lenox Ave & W 126 St	2	2020, 2023	53.44	83864	
Atlantic Ave & Fort Greene Pl	2	2023, 2025	49.42	125112	
Adam Clayton Powell Blvd & W 126 St	2	2023, 2025	46.15	76134	
6 Ave & W 34 St	2	2023, 2025	42.96	37507	
Hanson Pl & Ashland Pl	2	2020, 2023	41.93	149161	
Lenox Ave & W 117 St	2	2023, 2025	41.25	75237	
W 36 St & 9 Ave	2	2023, 2025	39.67	95215	
12 Ave & W 125 St	2	2020, 2025	39.04	70098	
DeKalb Ave & Hudson Ave	2	2023, 2025	39.00	76320	
E 43 St & 2 Ave	2	2020, 2023	38.74	47316	
W 35 St & 9 Ave	2	2023, 2025	36.74	96824	
Lispenard St & Broadway	2	2020, 2023	36.00	108742	
W 36 St & 7 Ave	2	2023, 2025	35.80	98388	
E 102 St & 1 Ave	2	2020, 2023	35.65	57103	
W 95 St & Broadway	2	2020, 2023	34.56	99835	
E 58 St & 1 Ave (NW Corner)	2	2020, 2023	33.92	45860	
31 St & Newtown Ave	2	2023, 2025	33.55	81540	
W 37 St & 5 Ave	2	2020, 2025	33.46	118954	
W 30 St & 8 Ave	2	2023, 2025	33.10	187201	
W 35 St & 8 Ave	2	2023, 2025	32.96	116572	
Broadway & W 36 St	2	2020, 2023	31.17	64169	
E 45 St & 3 Ave	2	2020, 2025	30.29	57698	
2 Ave & E 96 St	2	2020, 2023	28.58	104249	
W 43 St & 6 Ave	2	2023, 2025	28.27	65890	
Kenmare St & Elizabeth St	2	2023, 2025	27.97	158464	
S 5 Pl & S 5 St	2	2023, 2025	27.50	136999	
W 42 St & Dyer Ave	2	2023, 2025	26.90	151486	
E 51 St & 2 Ave	2	2020, 2025	25.73	105575	
W 52 St & 6 Ave	2	2020, 2025	25.49	96457	



EB VS WSI

Risk Seasonality (Median $\pm$ IQR) — mode=nyc (y2019_2020_2023_2025_m1_2_3_4_5_6_7_8_9_10_11_12_modenyc)



Risk Seasonality (Median $\pm$ IQR) — mode=nyc (y2019_2020_2023_2025_m1_2_3_4_5_6_7_8_9_10_11_12_modenyc)

